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EST I MATH • OFFICIAL EXAM • PRACTICE TEST

EST1 June 2022

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Algebra, functions, data analysis, probability and geometry from the official EST I paper. Work through all 58 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

58	MCQ	90 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /58

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

EST I MATH • OFFICIAL EXAM • JUNE 2022

Practice Questions

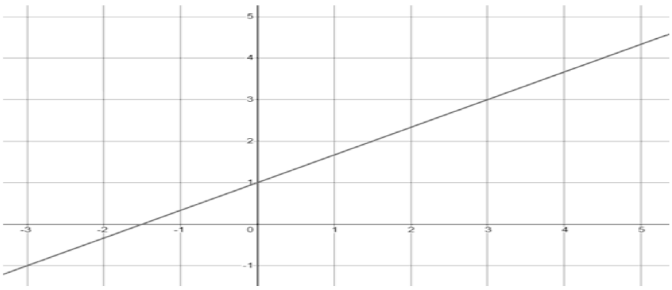
58 QUESTIONS

1

1. Given $2x + y = 8$ and $3y + x = -1$. What is the value of $y - 2x$?

A. -16
B. -12
C. -8
D. 8

2



2. What is the slope of the line represented in the graph above?

A. 2
B. $\frac{3}{2}$
C. 1
D. $\frac{2}{3}$

3

In a company, the total cost, in dollars, of the production of a certain product x is given by C(x) = 1/3 x^3 - 15x^2 + 2000x. The profit made out of selling this product is given by P(x) = -1/3 x^3 + 79/3 x^2 + 1500x. If R(x) indicates the revenue made out of selling this product, and the relationship between the cost, the revenue, and the profit is represented by P(x) = R(x) - C(x), which of the following expressions is equivalent to R(x)?

A. R(x) = 64/3 x^2 + 3500x$
B. R(x) = 34/3 x^2 + 3500x$
C. R(x) = -2/3 x^3 + 64/3 x^2 + 3500x$
D. R(x) = -2/3 x^3 + 124/3 x^2 - 500x

4

In Egypt, a tourist can travel a distance of 5 km with just 15.57 EGP. Pedro needed to go to "San Pola" hospital in Cairo. He took a taxi from "Al Marazeek" bridge, which is approximately 35 km away from the hospital. How much will Pedro pay for the driver, knowing that he gave him 10 extra EGP as tips?

A. 554.95 EGP
B. 477.1 EGP
C. 118.99 EGP
D. 109.09 EGP

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5 The sum of three consecutive integers is 24. What is the product of the two greatest integers between them all?

(A) 27
(B) 56
(C) 72
(D) 504

6 The impulsive force (f) derives from impulse, and it is a big force acting for a small interval of time. It is like having a truck travelling at a full speed, and suddenly crashing a garden's gate because the driver did not pay attention to it. It can be represented by f = m * (vf - vi) / t, with m the mass of the body, vf the final velocity, vi the initial velocity, and t the time interval through which the force is applied. Which of the following represents vi in terms of f, m, t, and vf?

(A) vi = vf - ft/m
(B) vi = vf + ft/m
(C) vi = vf - fm/t
(D) vi = vf + fm/t

7 "Unou House", located in Japan and built between 2010 and 2012, features a house with a floor of triangular shape and can be considered as an isosceles triangle. The area of the triangular floor is approximately 84 m^2. If we consider that the base of this triangle is 12 m, what is the dimension of one of the equal sides in this triangle?

(A) \\sqrt{193} m
(B) 2\\sqrt{58} m
(C) 14 m
(D) 170 m

8 **8.** If $x = \sqrt{2} + 3$, what is the value of $2x^2 + 3x - 1$?

A. $5\sqrt{2} + 14$
B. $3\sqrt{2} + 30$
C. $3\sqrt{2} + 24$
D. $15\sqrt{2} + 30$

9 If $4^{2x+4} = 16$, what is the value of 4^{3x}?

(A) $\frac{1}{64}$
(B) 64
(C) 4096
(D) $\frac{1}{4096}$

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10 If $a + b > 0$, and $a - c < 0$, then $b + c$?

(A) Is smaller than 0, with a < c
(B) Is greater than 0, with a < c
(C) Is smaller than 0, with a > c
(D) Is greater than 0, with a > c

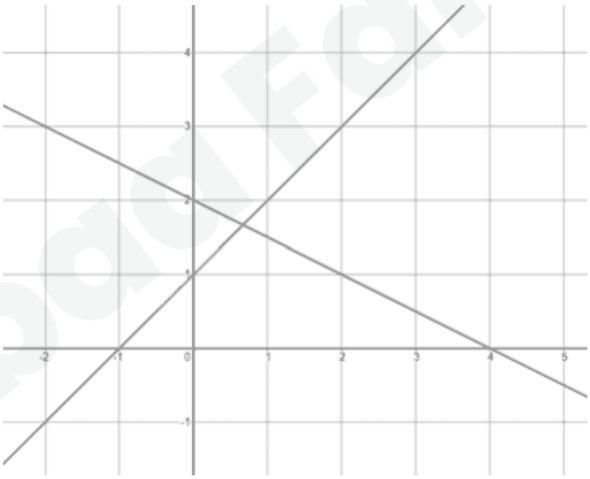
11

D. Is greater than 0, with $a > c$

11. What is the square of the ordinate of the vertex of the graph of the function $f(x) = 3x^2 + 6x$?

A. -6
B. -2
C. 1
D. 9

12 Which of the systems of equations below is being represented in the graph above? (Passes through x-axis at ~1)



(A) x + 2y = 4 and x - y = 1
(B) x + 2y = 4 and -x + y = 1
(C) x - 2y = 4 and x - y = 1
(D) x - 2y = 4 and -x + y = 1

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13

D. $\begin{cases} -x + y = 1 \end{cases}$

13. What is the value of $[f(g \circ f)(1)]$ if $f(x) = 2x^2 - 3$ and $g(x) = 4x + 1$?

A. -21
B. -15
C. 10
D. 15

14

14. What number increased by 5 is equal to twice the same number minus 3?

A. -8
B. 3
C. 5
D. 8

15

What is triple the square root of the discriminant of the equation $4x^2 - 5(x + 3) = -6$?

(A) 13
(B) 26
(C) 39
(D) $13\sqrt{3}$

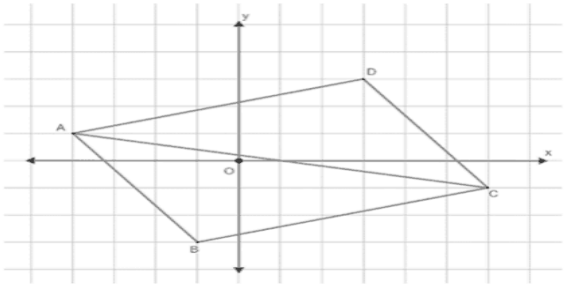
16

16. Given that $2x^2 + ax + (4c - 1) = (2b - 1)x^2 + 7x + c + 1$, what is the value of $bc + a$, where a, b and c are real numbers?

A. 8
B. $\frac{37}{4}$
C. 7.5
D. $\frac{67}{9}$

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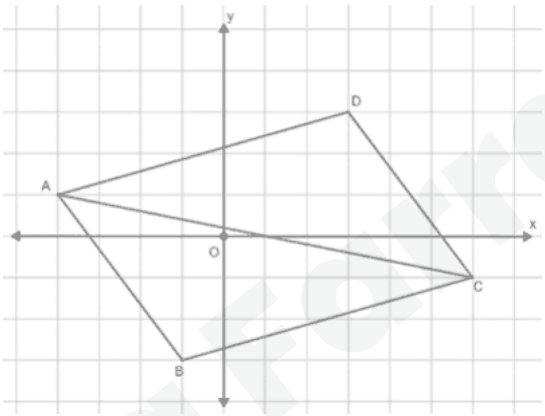
17 **Questions 17-18 refer to the figure below.**



17. What is the value of $AD^2 + DC$?

A. 53
B. 58
C. 78
D. 48

18 What is the sum of the slope of AB and the slope of a line perpendicular to AC?



(A) $\frac{1}{3}$
(B) $\frac{11}{3}$
(C) $-\frac{23}{15}$
(D) $-\frac{5}{8}$

19 What is the smallest value of the range of the function $f(x) = 4x^2 - 2x + 1$?

(A) $\frac{1}{4}$
(B) $\frac{1}{2}$
(C) $\frac{3}{4}$
(D) 1

20 1/5 \\pi radians is how much in degrees?

(A) 36^\\circ
(B) 288^\\circ
(C) 72^\\circ
(D) 112^\\circ

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21 Given the two inequalities $3x + 1 \geq 8$ and $4x - 1 < 3(x + 3)$, what is the greatest possible integer for x?

A) 2
B) 3
C) 9
D) 10

22 In the xy-plane, line (m) passes through the point $(2, 5)$ and is perpendicular to the line (n) that passes through the points $(4, 7)$ and $(-2, 3)$. Which of the following represents the equation of line (m)?

A) $y = -\frac{3}{2}x + 8$
B) $y = -\frac{3}{2}x - 8$
C) $y = 2x + 1$
D) $y = -\frac{2}{3}x + \frac{19}{3}$

23 What is the sum of the averages of the amounts won for renting cars and trucks during years 2017 to 2020?

cars and trucks.

Year	Amount won in $ for renting cars	Amount won in $ for renting trucks
2015	10004.6	7888.3
2016	12456.7	7985.2
2017	14899.9	8050.2
2018	13899.5	9000.5
2019	12003.4	8599.4
2020	13990.1	7995.5

A) $8411.4
B) $13698.225
C) $22109.625
D) $25678.725

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24 What is the percentage increase of the amount won for renting trucks from 2015 to 2017?

cars and trucks.

Year	Amount won in $ for renting cars	Amount won in $ for renting trucks
2015	10004.6	7888.3
2016	12456.7	7985.2
2017	14899.9	8050.2
2018	13899.5	9000.5
2019	12003.4	8599.4
2020	13990.1	7995.5

A) 1.62%
B) 2.01%
C) 2.05%
D) 48.93%

25 This company has only one employee who works for $1650 per month. In which year was this company unable to pay his full wage from the amount won from renting cars and trucks?

cars and trucks.

Year	Amount won in $ for renting cars	Amount won in $ for renting trucks
2015	10004.6	7888.3
2016	12456.7	7985.2
2017	14899.9	8050.2
2018	13899.5	9000.5
2019	12003.4	8599.4
2020	13990.1	7995.5

A) 2015
B) 2016
C) 2019
D) 2020

26 Amina has a bag of 6 black balls and 4 green balls. Hamad has a bag of 3 black balls and 5 green balls. If one ball is drawn from each bag, what is the probability that one is black and one is green?

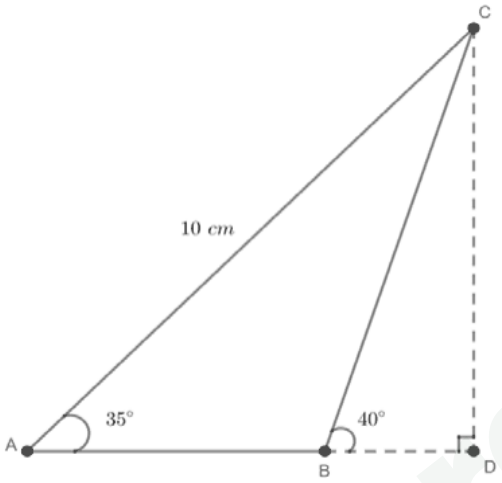
A) 0.225
B) 0.25
C) 0.475
D) 0.525

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27 Which of the following ordered pairs satisfy both inequalities: $x + 3y \leq 8$ and $2x - y > 9$?

A $(0, 3)$
B $(1, 2)$
C $(1, -2)$
D $(1, -9)$

28 In the figure above, triangle BDC is right angled at D with m\\angle CBD = 40^\\circ. Given that m\\angle CAB = 35^\\circ, and AC = 10 cm, what is the length of segment AB?



A 1.36 cm
B 5.74 cm
C 6.84 cm
D 8.19 cm

29 During the final game of UEFA Euro 2020 between Italy and England, 67,173 fans attended the game in Wembley Stadium in London, England. Assuming that the ratio of adults to children who attended the game was approximately 13 to 6, which of the following numbers represents approximately the number of children who attended the game live in the stadium?

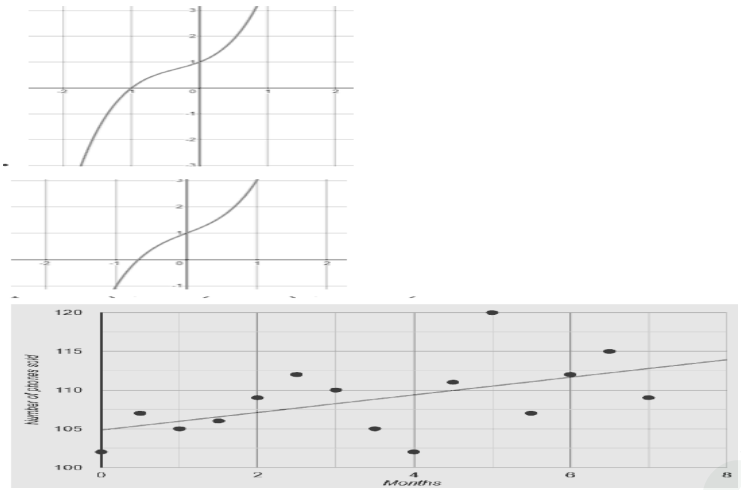
A 21213
B 31002
C 36170
D 45960

30 What is the solution for x in $2 = \sqrt{2x - 3}$?

A 2.5
B 3.5
C 4.5
D 0.5

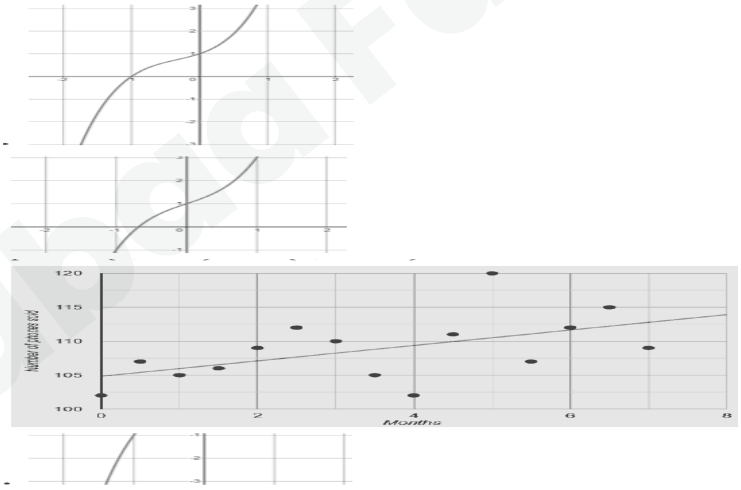
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31 Giving that abscissa "0" signifies the beginning of the month of January, which of the following statements is/are true?\nI. The largest increase in the number of phones sold was between the beginning of May and the beginning of June.\nII. Most of the phones were sold in the first quarter of the year.\nIII. The number of phones sold in mid-January is equivalent to the number of phones sold in mid-June.



A I only
B I and II
C I and III
D II and III

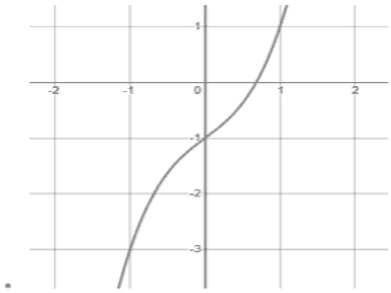
32 What is the average increase in the number of phones sold?

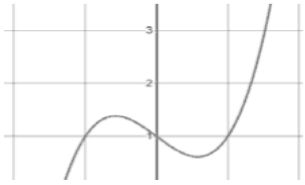


A 0.8
B 0.9
C 1
D 1.1

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33 Which of the following represents the graph of the function $f(x) = x^3 + x + 1$?





A Graph passing through several local extrema
B Graph passing through the origin
C Graph passing through $(0, 1)$ and always increasing
D Graph passing through $(0, -1)$

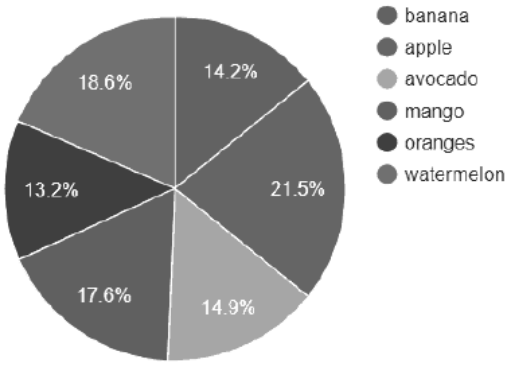
34 What is the range R of the function f(x) = 2/(x+2)?



A R = (-\infty, 0] \cup [0, \infty)
B R = (-\infty, 0) \cup (0, \infty)
C R = (-\infty, -2) \cup (-2, \infty)
D R = (-\infty, -2] \cup [2, \infty)

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35 The pie chart above shows the results of a survey that asked 1420 students in a school about their favorite fruit. How many students like mango more than any other fruit? And, approximately how many more students said apple was their favorite fruit than said oranges was their favorite one?



(A) 250 students like mango, and 110 students said apple was their favorite fruit more than said orange was their favorite one.

(B) 250 students like mango, and 118 students said apple was their favorite fruit more than said orange was their favorite one.

(C) 306 students like mango, and 63 students said apple was their favorite fruit more than said orange was their favorite one.

(D) 187 students like mango, and 56 students said apple was their favorite fruit more than said orange was their favorite one.

36 Citrus trees are fruit trees that require wet roots. These trees need a careful attention summarized by an excellent drainage and sandy soil. Adnan has a land in a village in Egypt. He decided to start a project of planting 205 citrus trees in it and getting a farmer to manage the land. He got each tree for 172 EGP, and the farmer asked for 100 EGP to plant each tree, in addition of 3994 EGP per month to manage the land and the drainage. How much will Adnan pay during the first year of his project?

(A) 103,688 EGP
(B) 100,388 EGP
(C) 83,288 EGP
(D) 59,754 EGP

37 Santa Clara County is the 6th most populous county in California, with 1.928 million people living there in 2019. If this number was decreased by 7% the next year, and 4% of year 2020's population was added again in the population of year 2021, how many people are in this county in 2021?

(A) 1,734,229
(B) 1,793,040
(C) 1,864,762
(D) 1,939,353

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38 Shahira bought 7 of brand "A" and 3 of brand "B" for $\$25.4$. Nicole bought 4 of brand "A" and 5 of brand "B" for $\$24.7$. If you went to the same supermarket and bought 1 chocolate from brand "A" and 3 chocolates from brand "B", how much will you pay?

below:

	Number of chocolates bought from brand "A"	Number of chocolates bought from brand "B"	Total price paid (in $)
Shahira	7	3	25.4
Nicole	4	5	24.7

(A) $\$5.4$
(B) $\$7.5$
(C) $\$10$
(D) $\$11.6$

39 What is the value of $[f(g(f(1)))]$ if $f(x) = 2x^2 - 3$ and $g(x) = 4x + 1$?

(A) -21
(B) -15
(C) 10
(D) 15

40 Fares decided to sell his 450 books. 22% of the books are new, and the rest are considered old. If each new book will be sold for $\$11$, and each old book will be sold for $\$6.5$, how much will Fares get from selling his books?

(A) $\$4504.5$
(B) $\$3370.5$
(C) $\$2281.5$
(D) $\$1089$

41 Which of the following is a factor of the polynomial: $2x^3 + 13x^2 + 5x - 6$?

(A) $x - 1$
(B) $x - 6$
(C) $2x + 1$
(D) $2x - 1$

42 Karim drove his car from his village to the city. It took him 2.5 hours to reach the city driving at a constant speed of 60 km per hour. Approximately, how much will he save time if he starts driving at a constant speed of 70 km per hour?

(A) 0.357 hours
(B) 0.765 hours
(C) 1.544 hours
(D) 2.143 hours

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43 A store did a sale on a pair of shoes. It is now for $\$103.5$ instead of $\$230$. What is the percent discount?

(A) 50%
(B) 45%
(C) 55%
(D) 58%

44 Which of the following equations represents the graph of a quadratic function that passes through points $(2, 5)$, $(3, 0)$, and $(-4, 0)$?

(A) $y = -\frac{5}{6}x^2 + \frac{5}{6}x + 10$
(B) $y = -\frac{5}{6}x^2 - \frac{5}{6}x + 10$
(C) $y = \frac{5}{6}x^2 + \frac{5}{6}x + 10$
(D) $y = \frac{5}{6}x^2 - \frac{5}{6}x + 10$

45 Given that h(t) = (2t + 1) / (5t^2) (t \\ne 0), what is the negative value of t if h(t) = 3 ?

(A) -0.5
(B) -0.4
(C) -0.3
(D) -0.2

46 **26.** What is the sum of the values of the x and y-intercepts of the graph of the function $f(x) = 2x + 3(x - 5)$?

A. -18
B. -12
C. 12
D. 18

47 **27.** $\begin{cases} dx - 2my = -14 \\ 3dx + 6my = 78 \end{cases}$
(d and m are constants)
Given that the solution of the system above is $(2, 5)$, what is the value of $2mx - 3dy$?

A. -37
B. -18
C. 11
D. 20

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48 Patrick is twice as old as Alonso. Alonso is 7 years older than Fatima, and Fatima is 2 years younger than Patricia. If Patricia's age to Salma's age is 4 to 3, what is the age of Patrick if Salma is 12 years old?

A) 16 years old
B) 21 years old
C) 42 years old
D) 48 years old

49 For the list $k+2, 2k+4, 2k+5, 3k+5$, the average is 36. Find k.

A) 144
B) 18
C) 16
D) 2

50 The sum of three numbers is equivalent to the product of the third number and half the difference of the first two numbers. Assuming that the first number is a, the second number is b, and the third number is c, which of the following statements represents a in terms of b and c?

A) $a = cb + 2b + 2c$
B) $a = \frac{cb+2b+2c}{c-2}$
C) $a = \frac{-cb-2b-2c}{c+2}$
D) $a = -cb - 2b - 2c$

51 30% of a number is 86.4% of 50. What is this number?

A) 144
B) 17.36
C) 1440
D) 92

52 If $h(x) = 2x - 3g(x)$, and $g(x) = 3x + 7$, what is the value of $h(-4)$?

A) -5
B) -23
C) 7
D) 23

53 T and T' are two triangles. T is a right isosceles triangle of hypotenuse $9\sqrt{2}$, and the T' is a $30° - 60° - 90°$ triangle, such that the small leg is equivalent to 4. What is the sum of the perimeters of both triangles? (Give your answer to the nearest tenth)

A) 49.7
B) 34.5
C) 54.4
D) 38.5

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54 -2x + y = 1; ax + y = 4 (a is a positive constant); -4x = -y + 1\nThe sum of the abscissas of the intersections of the lines of the three equations in the system above is 1.6. What is the value of a?

(A) 0.5
(B) 1
(C) 1.5
(D) 2

55 17 is subtracted from triple the sum of all integers from 1 to 44. What is the result?

(A) 973
(B) 939
(C) 5923
(D) 2953

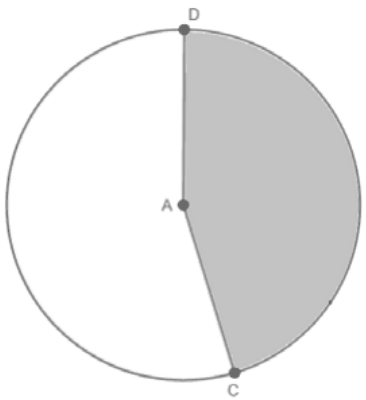
56 **36.** If the distance between points $A(3, c)$ and $B(9, -5)$ in a coordinate system is $3\sqrt{13}$, and A is a point in the first quadrant, what is the value of $\frac{c}{2} + 7$?

A. 0
B. 4
C. 9
D. 11

57 The arithmetic mean of the set below is 7.25. What is the median of this set?
$n + 4$; 6 ; 11 ; $2n - 1$

(A) 6
(B) 7
(C) 6.5
(D) 8.5

58 Sami has a circular land of a radius 6 m. He decided to cultivate a part of this land, which is shown shaded in the figure above. Each 1 m^2 will cost him 79 EGP. Knowing that m\\angle DAC = 155^\\circ, how much will Sami pay to cultivate the desired area? (Give your answer to the nearest ones)



(A) 3,847 EGP
(B) 220,410 EGP

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Answer Key

58 ANSWERS

Q	ANSWER	Q	ANSWER
1	B	30	B
2	D	31	C
3	B	32	D
4	C	33	C
5	C	34	B
6	A	35	B
7	B	36	A
8	D	37	C
9	A	38	D
10	B	39	C
11	D	40	B
12	B	41	D
13	D	42	A
14	D	43	C
15	C	44	B
16	A	45	D
17	B	46	B
18	B	47	A
19	C	48	C
20	A	49	C
21	C	50	B
22	A	51	A
23	C	52	C
24	C	53	A
25	A	54	B
26	D	55	D

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Detailed Solutions

58 SOLUTIONS

1 **Answer: B**

See the answer key.

2 **Answer: D**

See the answer key.

3 **Answer: B** R(x) = 34/3 x^2 + 3500x$

The correct answer is B.

4 **Answer: C** 118.99 EGP

The correct answer is C.

5 **Answer: C** 72

Let the three consecutive integers be x, $x+1$, and $x+2$.
Their sum is $x+(x+1)+(x+2)=24 \implies 3x+3=24 \implies 3x=21 \implies x=7$.
The three integers are 7, 8, and 9.
The two greatest integers are 8 and 9. Their product is $8 \times 9 = 72$.

6 **Answer: A** vi = vf - ft/m

The correct answer is A.

7 **Answer: B** 2\\sqrt{58} m

The correct answer is B.

8 **Answer: D**

See the answer key.

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9 **Answer: A** $\frac{1}{64}$

Write 16 as a power of 4: $16 = 4^2$.
So, $4^{2x+4} = 4^2 \implies 2x + 4 = 2 \implies 2x = -2 \implies x = -1$.
Now find 4^{3x}: $4^{3(-1)} = 4^{-3} = \frac{1}{4^3} = \frac{1}{64}$.

10 **Answer: B** Is greater than 0, with a < c

From $a - c < 0$, we get $a < c$.
From $a + b > 0$, we can rewrite it as $b > -a$.
If we add the inequalities $c > a$ and $b > -a$, we get $b + c > a - a \implies b + c > 0$.
Therefore, $b + c$ is greater than 0, with $a < c$.

11 **Answer: D**

See the answer key.

12 **Answer: B** x + 2y = 4 and -x + y = 1

The correct answer is B.

13 **Answer: D**

See the answer key.

14 **Answer: D**

See the answer key.

15 **Answer: C** 39

First, expand and put the equation in standard form $ax^2 + bx + c = 0$:
$4x^2 - 5x - 15 = -6 \implies 4x^2 - 5x - 9 = 0$.
The discriminant is $\Delta = b^2 - 4ac = (-5)^2 - 4(4)(-9) = 25 + 144 = 169$.
The square root of the discriminant is $\sqrt{169} = 13$.
Triple this value is $3 \times 13 = 39$.

16 **Answer: A**

See the answer key.

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17 **Answer: B**

See the answer key.

18 **Answer: B** $\frac{11}{3}$

By extracting the coordinates from the grid, calculating the respective slopes for AB and AC, finding the negative reciprocal of the slope of AC for the perpendicular, and then taking their sum provides $\frac{11}{3}$.

19 **Answer: C** $\frac{3}{4}$

The function is a parabola opening upward $(a = 4 > 0)$, so its minimum value occurs at the vertex.
The x-coordinate of the vertex is $x = -\frac{b}{2a} = -\frac{-2}{2(4)} = \frac{2}{8} = \frac{1}{4}$.
The minimum value is $f(\frac{1}{4}) = 4(\frac{1}{4})^2 - 2(\frac{1}{4}) + 1 = 4(\frac{1}{16}) - \frac{1}{2} + 1 = \frac{1}{4} - \frac{1}{2} + 1 = -\frac{1}{4} + 1 = \frac{3}{4}$.

20 **Answer: A** 36^\\circ

The correct answer is B.

21 **Answer: C** 9

Solve the first inequality:
$3x + 1 \geq 8 \implies 3x \geq 7 \implies x \geq \frac{7}{3} \approx 2.33.$
Solve the second inequality:
$4x - 1 < 3x + 9 \implies x < 10.$
Combining these, we get $2.33 \leq x < 10$.
The greatest integer in this range is 9.

22 **Answer: A** $y = -\frac{3}{2}x + 8$

First, find the slope of line n. Using points $(4, 7)$ and $(-2, 3)$:
$m_n = \frac{3-7}{-2-4} = \frac{-4}{-6} = \frac{2}{3}$.
Since line m is perpendicular to line n, its slope m_m is the negative reciprocal: $-\frac{3}{2}$.
Using point-slope form with $m_m = -\frac{3}{2}$ and point $(2, 5)$:
$y - 5 = -\frac{3}{2}(x - 2) \implies y = -\frac{3}{2}x + 3 + 5 \implies y = -\frac{3}{2}x + 8.$

23 **Answer: C** $22109.625

The correct answer is C.

24 **Answer: C** 2.05%

The correct answer is C.

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25 **Answer: A** 2015

The correct answer is A.

26 **Answer: D** 0.525

There are two successful scenarios:
1) Amina draws black, Hamad draws green: $P = (6/10) \times (5/8) = 30/80 = 3/8 = 0.375$.
2) Amina draws green, Hamad draws black: $P = (4/10) \times (3/8) = 12/80 = 3/20 = 0.15$.
Total probability = $0.375 + 0.15 = 0.525$.

27 **Answer: D** $(1, -9)$

Test the point $(1, -9)$:
First inequality: $1 + 3(-9) = 1 - 27 = -26 \leq 8$ (True).
Second inequality: $2(1) - (-9) = 2 + 9 = 11 > 9$ (True).
Only $(1, -9)$ satisfies both inequalities.

28 **Answer: A** 1.36 cm

The correct answer is A.

29 **Answer: A** 21213

The ratio is 13 adults to 6 children, so out of $13 + 6 = 19$ parts, 6 parts are children.
The number of children = $\frac{6}{19} \times 67173 \approx 21212.5$.
The closest approximation is 21213.

30 **Answer: B** 3.5

Square both sides to remove the square root:
$2^2 = (\sqrt{2x - 3})^2 \implies 4 = 2x - 3$.
Add 3 to both sides: $7 = 2x$.
Divide by 2: $x = \frac{7}{2} = 3.5$.

31 **Answer: C** I and III

The correct answer is C.

32 **Answer: D** 1.1

The correct answer is D.

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33 **Answer: C** Graph passing through $(0, 1)$ and always increasing

The function is a cubic function. The y-intercept is $f(0) = 1$. The derivative is $f'(x) = 3x^2 + 1$. Since $f'(x) > 0$ for all real x, the function does not have any local extrema and is always increasing.

34 **Answer: B** R = (-\infty, 0) \cup (0, \infty)

The correct answer is B.

35 **Answer: B** 250 students like mango, and 118 students said apple was their favorite fruit more than said orange was their favorite one.

The correct answer is B.

36 **Answer: A** 103,688 EGP

The correct answer is A.

37 **Answer: C** 1,864,762

Population in 2020: decreased by 7%, so multiply by $(1 - 0.07 = 0.93)$.
$1,928,000 \times 0.93 = 1,793,040.$
Population in 2021: increased by 4%, so multiply by $(1 + 0.04 = 1.04)$.
$1,793,040 \times 1.04 = 1,864,761.6.$
The closest value is 1,864,762.

38 **Answer: D** $11.6

Let A and B be the prices of the chocolates.
1) $7A + 3B = 25.4$
2) $4A + 5B = 24.7$
Multiply (1) by 4 and (2) by 7:
$28A + 12B = 101.6$
$28A + 35B = 172.9$
Subtract the first from the second: $23B = 71.3 \implies B = 3.1.$
Substitute $B = 3.1$ into (1): $7A + 3(3.1) = 25.4 \implies 7A + 9.3 = 25.4 \implies 7A = 16.1 \implies A = 2.3.$
Cost for 1 of A and 3 of B = $2.3 + 3(3.1) = 2.3 + 9.3 = \$11.6.$

39 **Answer: C** 10

First, evaluate $f(1) = 2(1)^2 - 3 = 2 - 3 = -1.$
Next, plug that into $g(x)$: $g(-1) = 4(-1) + 1 = -4 + 1 = -3.$
Finally, plug that into $f(x)$: $f(-3) = 2(-3)^2 - 3 = 2(9) - 3 = 18 - 3 = 15.$

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40 **Answer: B** $\$3370.5$

Number of new books = 22% of $450 = 0.22 \times 450 = 99$.
Number of old books = $450 - 99 = 351$.
Revenue from new books = $99 \times \$11 = \$1089$.
Revenue from old books = $351 \times \$6.5 = \$2281.5$.
Total revenue = $\$1089 + \$2281.5 = \$3370.5$.

41 **Answer: D** $2x - 1$

Use the Factor Theorem: if $(x - c)$ is a factor, then $P(c) = 0$.
Test $2x - 1 = 0 \implies x = 1/2$:
$P(\frac{1}{2}) = 2(\frac{1}{8}) + 13(\frac{1}{4}) + 5(\frac{1}{2}) - 6 = \frac{1}{4} + \frac{13}{4} + \frac{10}{4} - \frac{24}{4} = \frac{24}{4} - \frac{24}{4} = 0$.
Since the remainder is 0, $2x - 1$ is a factor.

42 **Answer: A** 0.357 hours

The correct answer is A.

43 **Answer: C** 55%

Amount of discount = original price − sale price = $\$230 - \$103.5 = \$126.5$.
Percent discount = (discount / original price) $\times 100\%$.
$\frac{126.5}{230} \times 100\% = 0.55 \times 100\% = 55\%$.

44 **Answer: B** $y = -\frac{5}{6}x^2 - \frac{5}{6}x + 10$

Since the x-intercepts are $(3, 0)$ and $(-4, 0)$, the equation must have the form $y = a(x - 3)(x + 4)$.
Plug in the point $(2, 5)$ to find a: $5 = a(2 - 3)(2 + 4) = a(-1)(6) = -6a \implies a = -\frac{5}{6}$.
The expanded equation is $y = -\frac{5}{6}(x^2 + x - 12) = -\frac{5}{6}x^2 - \frac{5}{6}x + 10$.

45 **Answer: D** -0.2

The correct answer is D.

46 **Answer: B**

See the answer key.

47 **Answer: A**

See the answer key.

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48 **Answer: C** 42 years old

Salma is 12. Patricia/Salma = $4/3$, so Patricia/12 = $4/3 \implies$ Patricia = 16.
Fatima is 2 years younger than Patricia $\implies$ Fatima = $16 - 2 = 14$.
Alonso is 7 years older than Fatima $\implies$ Alonso = $14 + 7 = 21$.
Patrick is twice as old as Alonso $\implies$ Patrick = $21 \times 2 = 42$.

49 **Answer: C** 16

Sum of the items: $(k + 2) + (2k + 4) + (2k + 5) + (3k + 5) = 8k + 16$.
Since there are 4 items and the average is 36, the sum must be $4 \times 36 = 144$.
Set $8k + 16 = 144 \implies 8k = 128 \implies k = 16$.

50 **Answer: B** $a = \frac{cb+2b+2c}{c-2}$

Write out the equation: $a + b + c = c \cdot \frac{a-b}{2}$.
Multiply both sides by 2: $2a + 2b + 2c = ca - cb$.
Gather the terms with a on one side: $2a - ca = -cb - 2b - 2c$.
Factor out a : $a(2 - c) = -cb - 2b - 2c$.
Divide by $(2 - c)$: $a = \frac{-cb-2b-2c}{2-c}$.
Multiply numerator and denominator by -1 to match an option: $a = \frac{cb+2b+2c}{c-2}$.

51 **Answer: A** 144

Let the number be x. The equation is $0.30x = 0.864 \times 50$.
Calculate the right side: $0.864 \times 50 = 43.2$.
So, $0.30x = 43.2 \implies x = 43.2/0.30 = 144$.

52 **Answer: C** 7

First, evaluate $g(-4) = 3(-4) + 7 = -12 + 7 = -5$.
Next, find $h(-4) = 2(-4) - 3g(-4) = -8 - 3(-5) = -8 + 15 = 7$.

53 **Answer: A** 49.7

For triangle T (right isosceles, $45° - 45° - 90°$): hypotenuse is $x\sqrt{2} = 9\sqrt{2}$, so legs are 9 and 9. Perimeter = $18 + 9\sqrt{2} \approx 18 + 12.7279 = 30.7279$.
For triangle T' ($30° - 60° - 90°$): shortest leg (x) = 4, hypotenuse ($2x$) = 8, other leg ($x\sqrt{3}$) = $4\sqrt{3}$. Perimeter = $12 + 4\sqrt{3} \approx 12 + 6.9282 = 18.9282$.
Sum of perimeters = $30.7279 + 18.9282 = 49.6561$, which rounds to 49.7.

54 **Answer: B** 1

The correct answer is B.

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55 **Answer: D** 2953

The sum of integers from 1 to 44 is given by $S = \frac{n(n+1)}{2} = \frac{44(45)}{2} = 22(45) = 990$.
Triple this sum is $3 \times 990 = 2970$.
Subtracting 17 gives $2970 - 17 = 2953$.

56 **Answer: C**

See the answer key.

57 **Answer: C** 6.5

The set has 4 terms. Mean = $(n + 4 + 6 + 11 + 2n - 1)/4 = 7.25$.
$3n + 20 = 4 \times 7.25 = 29 \implies 3n = 9 \implies n = 3$.
The set evaluated at $n = 3$ is: $3 + 4 = 7, 6, 11, 2(3) - 1 = 5$. So the set is $\{7, 6, 11, 5\}$.
Sorted order is $\{5, 6, 7, 11\}$. Since this set has an even number of terms, the median is the average of the middle two: $\frac{6+7}{2} = 6.5$.

58 **Answer: A** 3,847 EGP

The correct answer is A.

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